Project Initialization and Planning Phase

Date	28 June 2026
Team ID	xxxxx
Project Title	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to develop an AI-powered **Emotion Detection & Learning Support Engine** that analyzes students' text input, detects emotions using **BiLSTM** and **BERT** models, and generates personalized, empathetic learning guidance through **Google Gemini AI**.
Scope	The project preprocesses student text, performs emotion classification, detects mixed emotions, generates AI-powered learning guidance, visualizes analytics, and stores session history for future reference.
Problem Statement	
Description	Many students struggle to understand difficult concepts because existing learning platforms provide generic responses without considering their emotional state. This often leads to confusion, frustration, and reduced learning motivation.
Impact	The proposed solution improves student engagement, learning confidence, academic performance, and provides personalized emotional and educational support.
Proposed Solution	
Approach	Develop an AI-powered learning support system using **Python, Streamlit, TensorFlow (BiLSTM), Hugging Face BERT, Natural Language Processing (NLP),** and **Google Gemini AI** to analyze student text, detect emotions, and generate personalized learning guidance.
Key Features	• Emotion detection using **BiLSTM** and **BERT** models. • Text preprocessing and keyword enhancement. • Mixed emotion detection with confidence scoring. • AI-powered personalized learning guidance using **Google Gemini**

	AI. • Interactive Streamlit user interface. • Emotion analytics dashboard and visualization. • Session history and CSV-based record management

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 Processor or higher
Memory	RAM specifications	8 GB or higher
Storage	Disk space for data, models, and logs	10 GB Free SSD Storage
Software		
Frameworks	Python frameworks	Python, Streamlit, TensorFlow
Libraries	AI/ML & NLP Libraries	Transformers, TensorFlow, NLTK, Pandas, NumPy, Plotly, Joblib, Scikit-learn
Development Environment	IDE	Visual Studio Code
Data		
Data	Source, size, format	GoEmotions, EmpatheticDialogues, ISEAR Dataset, Emotion Text Dataset (CSV), Processed Dataset, Tokenizer (.pkl), Label Encoder (.pkl)